

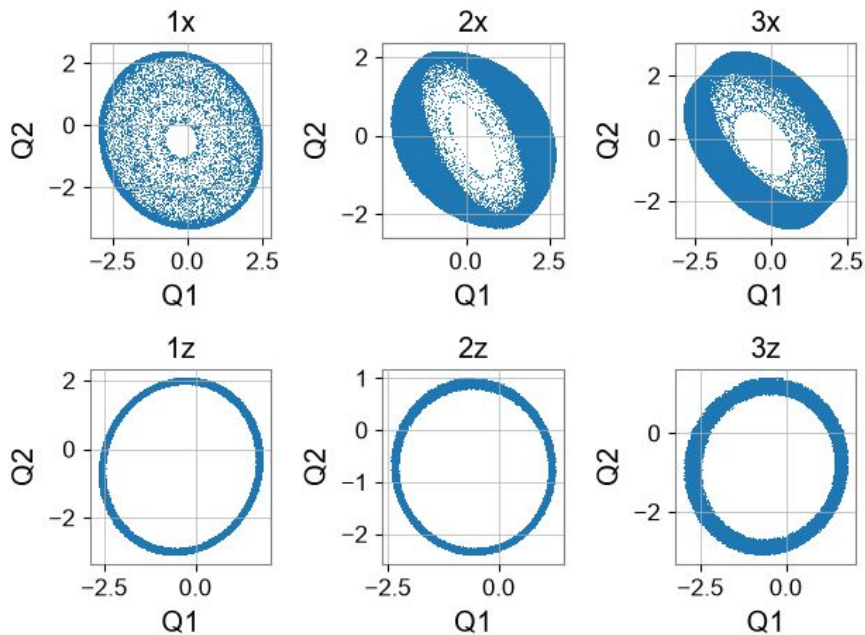
WELCOME TO THE INTERN PRESENTATION

(annoying interns)

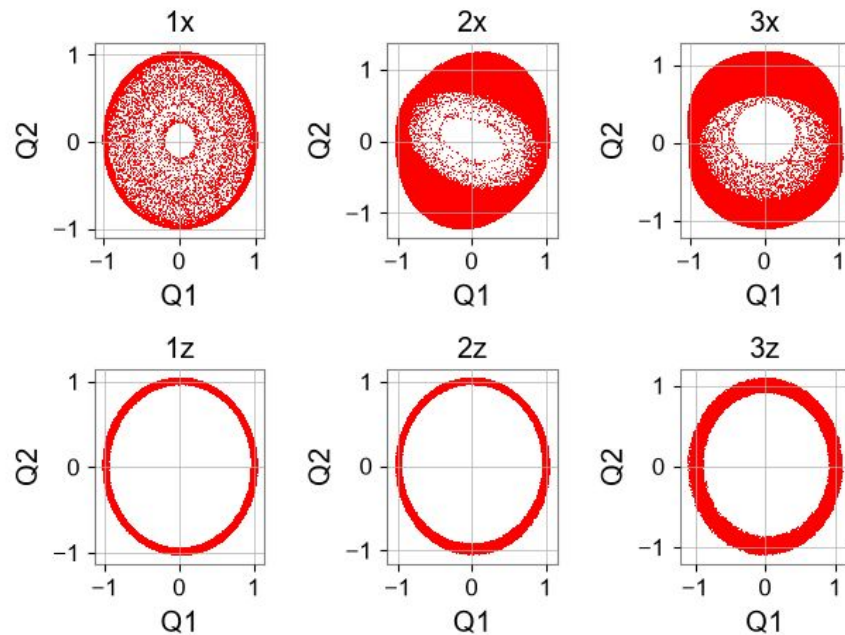




Ellipse fitting



Non-fitted

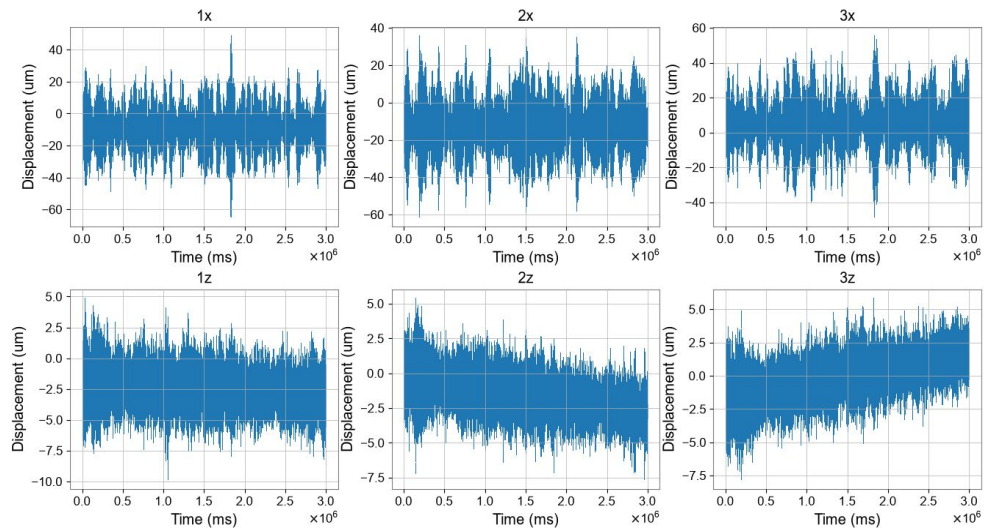
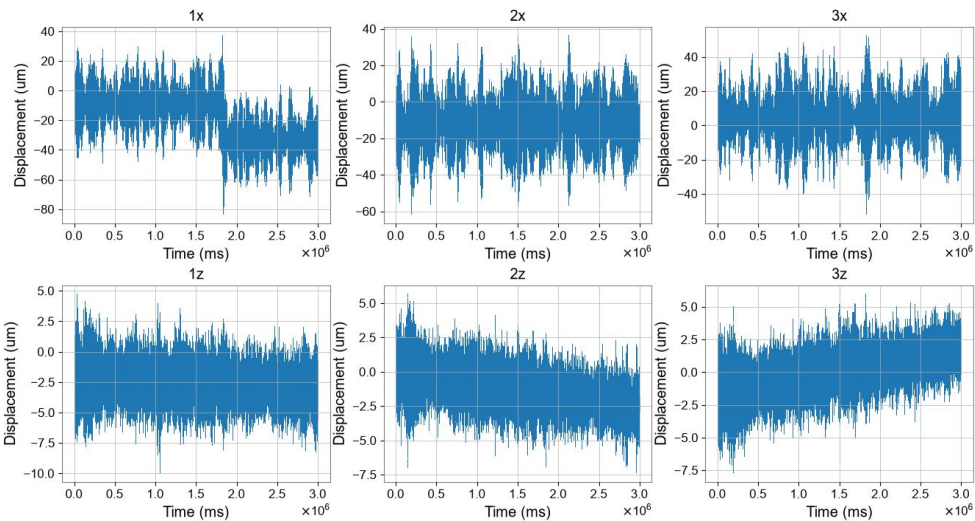


Single ellipse fitted

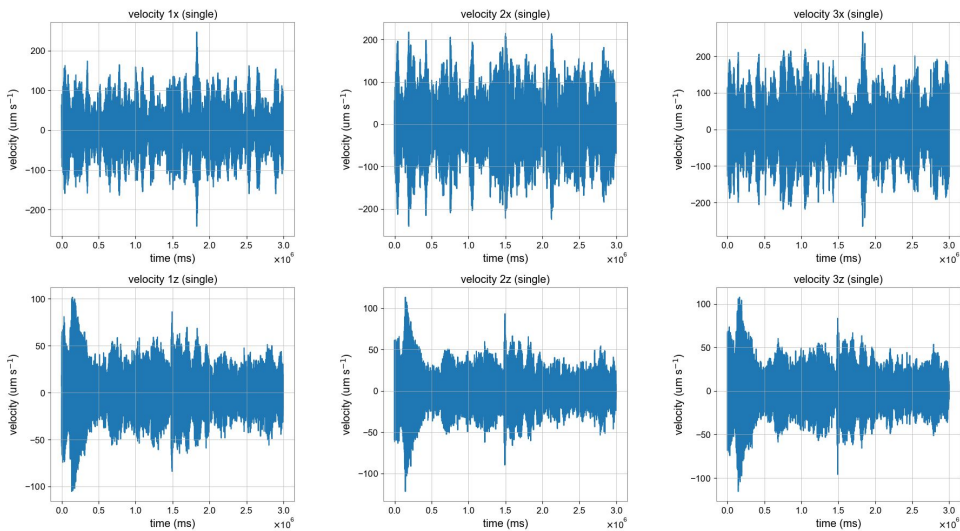


Displacement time series

Non-fitted



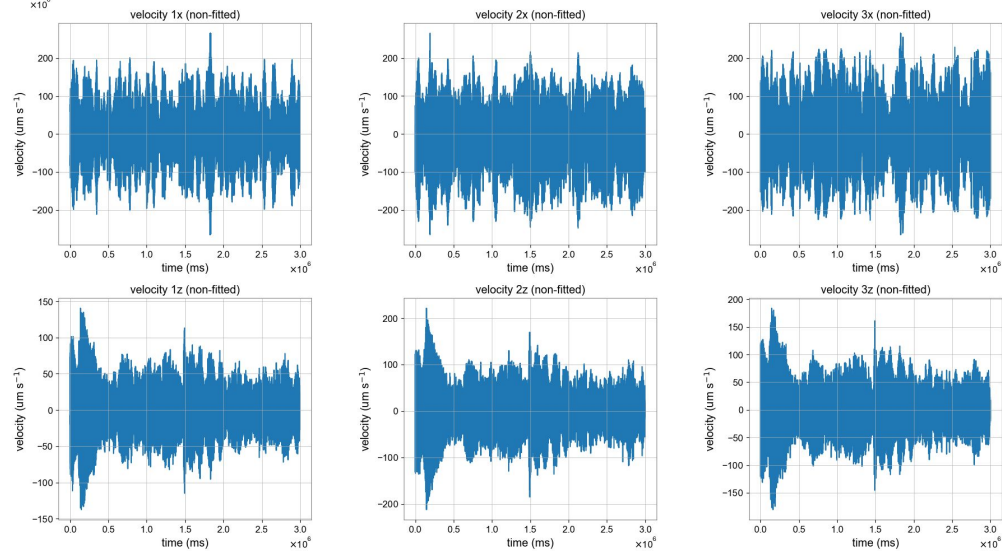
Single ellipse fitted



Single ellipse fitted

Non-fitted

**Velocity
time series**





Post processing

- **Local** fitting using windows
- More **accurate** transformations



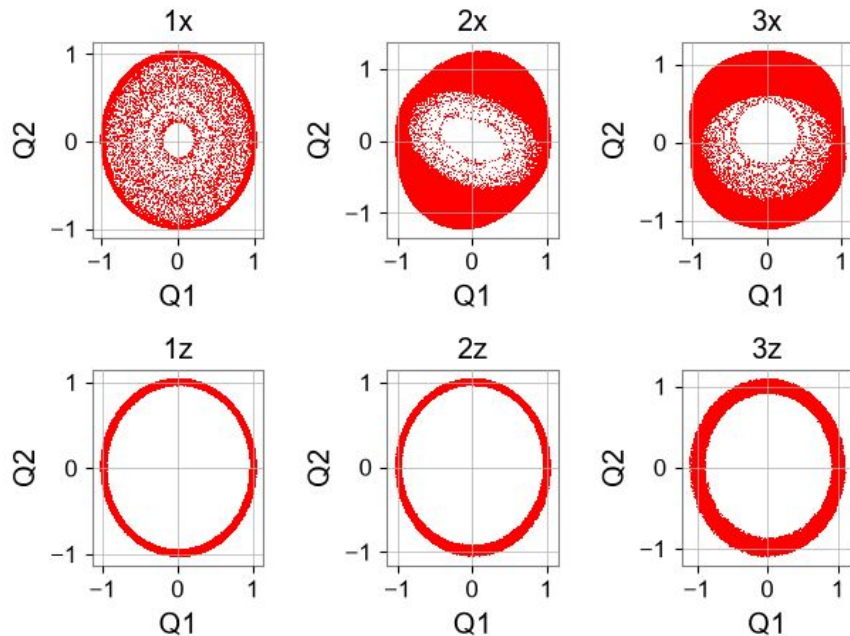
Real-time processing

- **Predicting** the ellipse parameters
 - More **quiet** dataset
- **RBF** model
- **kNN** algorithm (machine learning model)

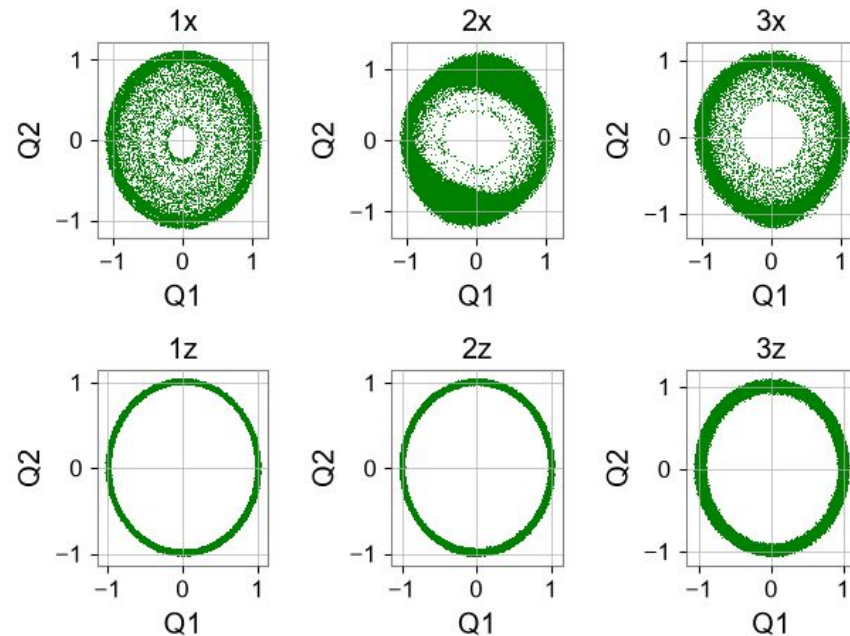




Post processing

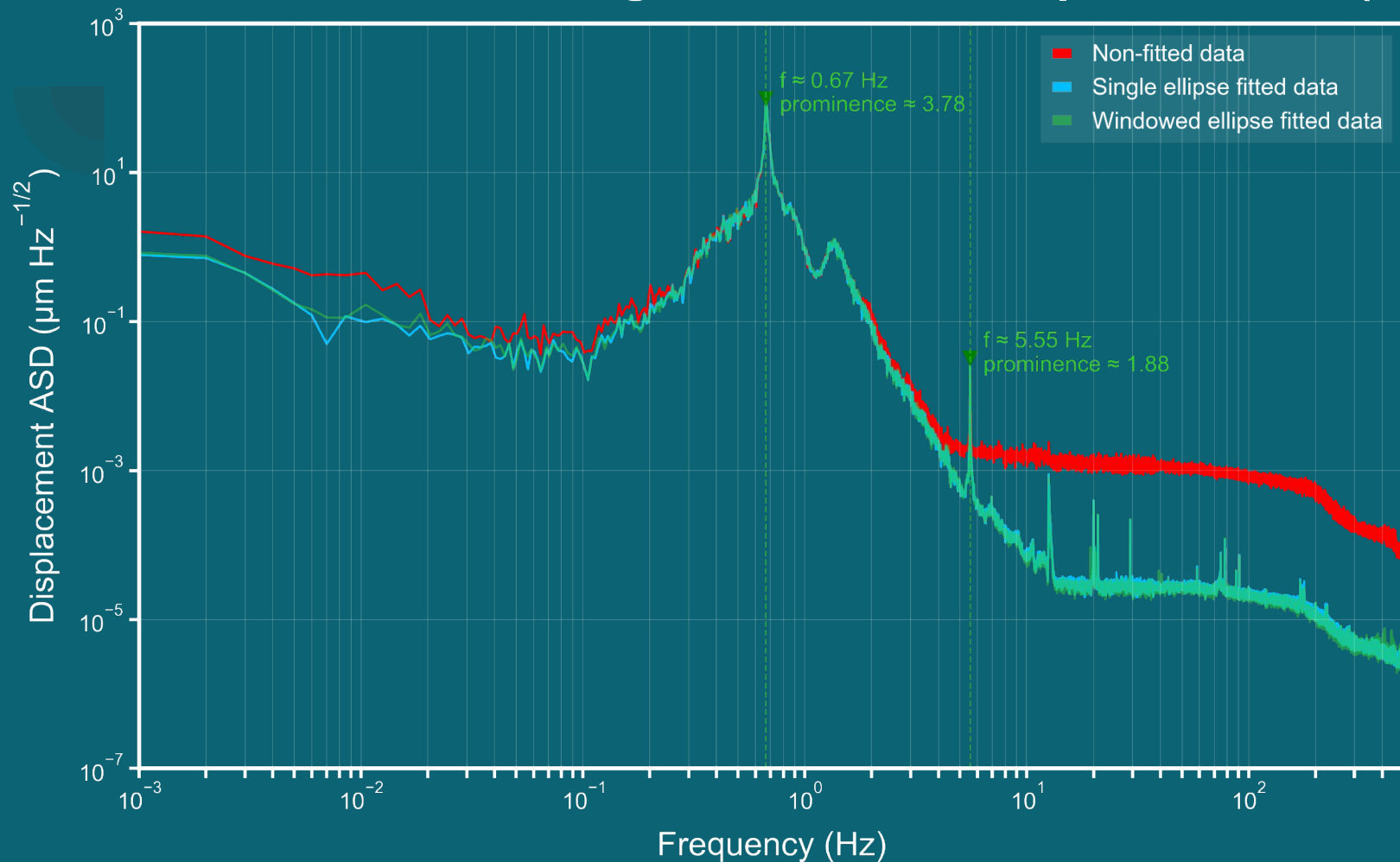


Single ellipse fitted

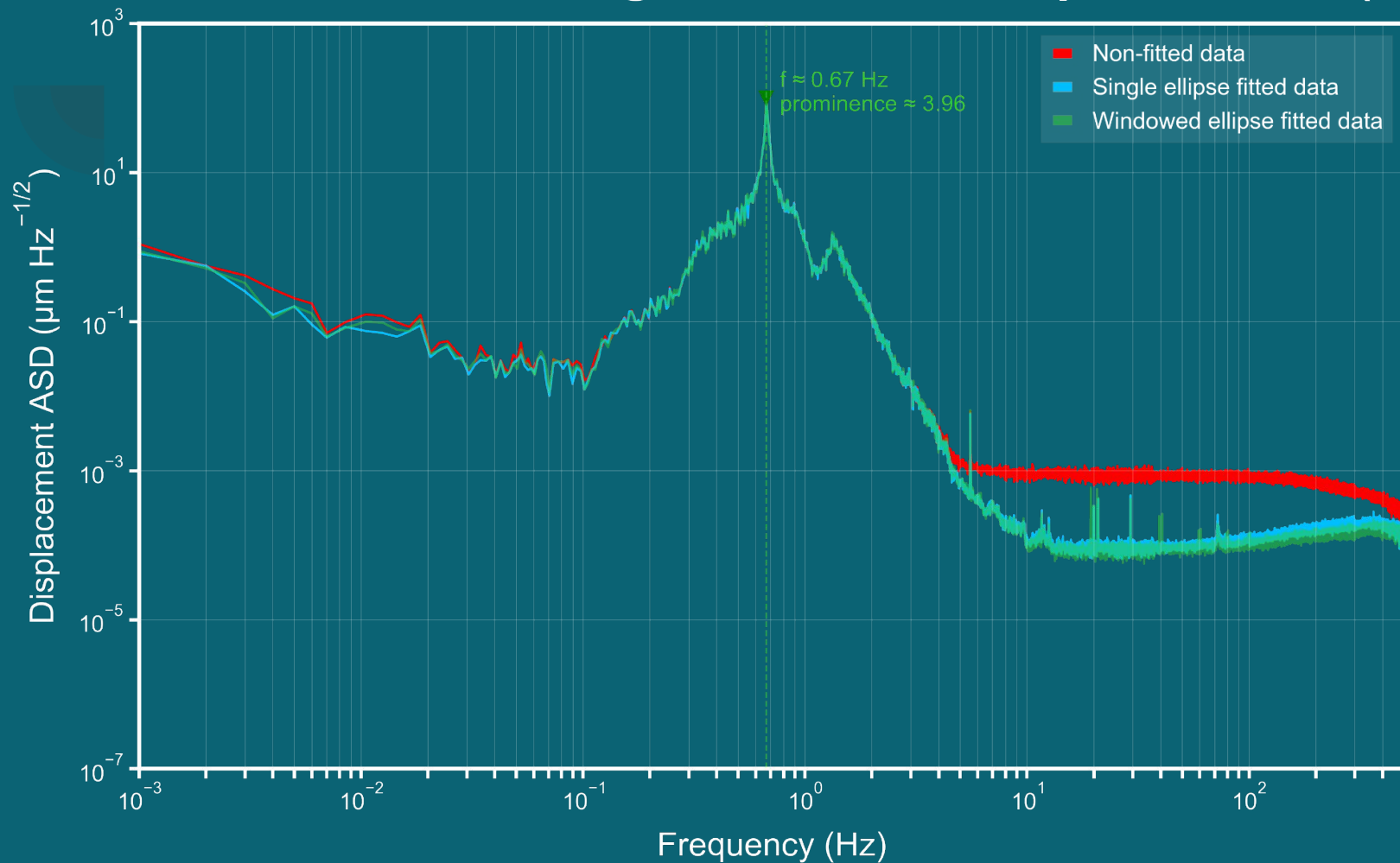


Windowed ellipse fitted

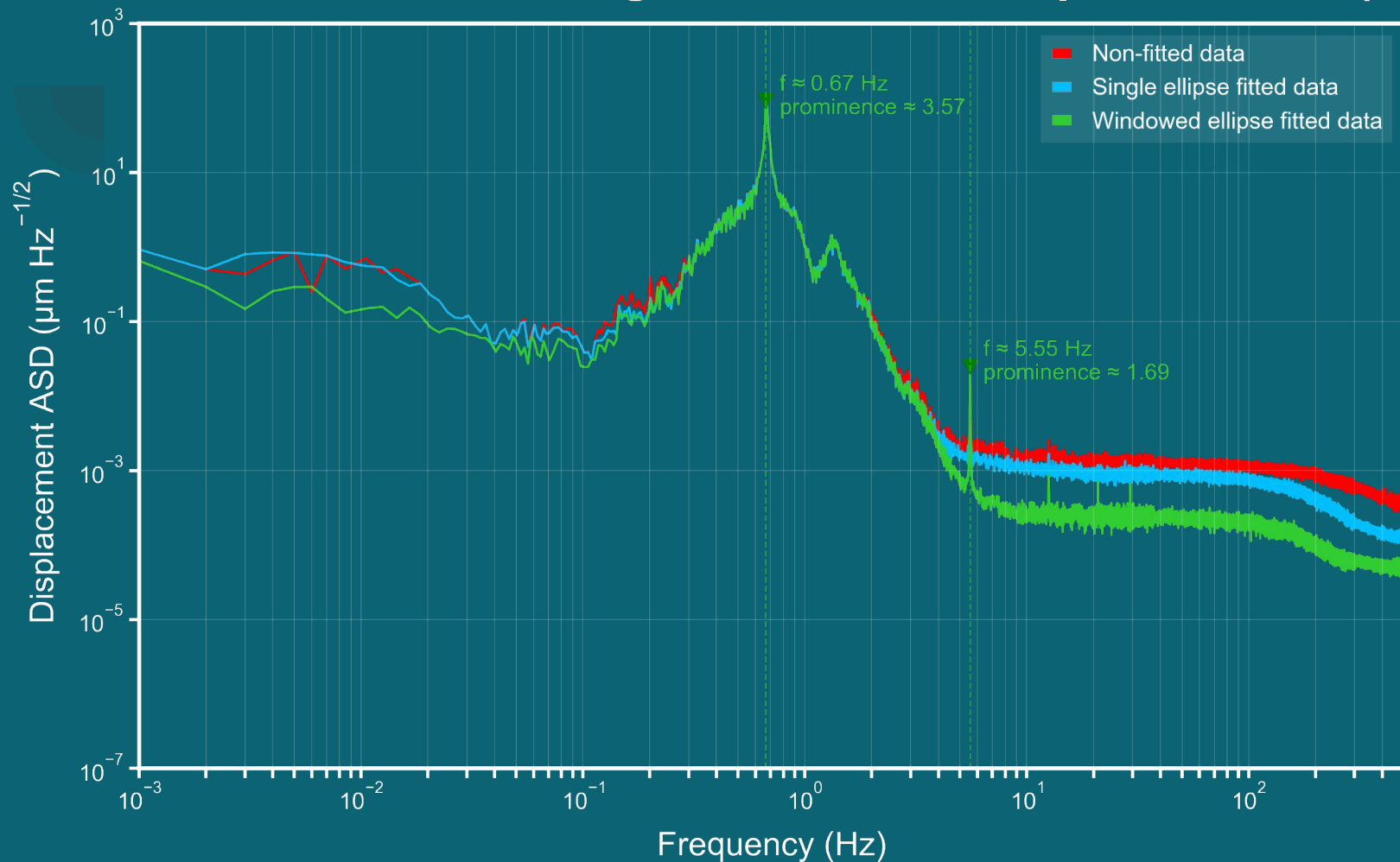
Smoothened ASD diagram for HoQI displacements (1x)



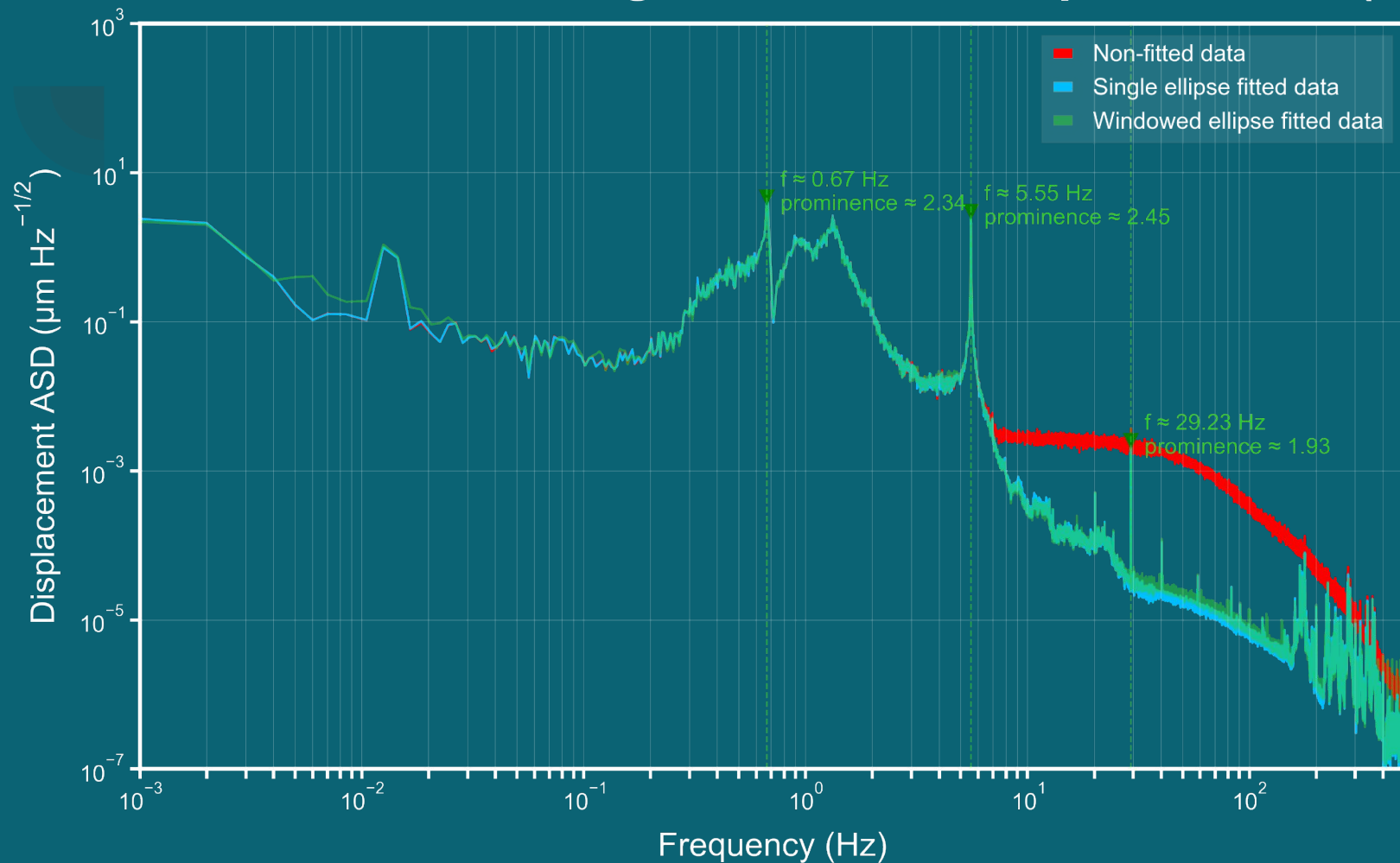
Smoothened ASD diagram for HoQI displacements (2x)



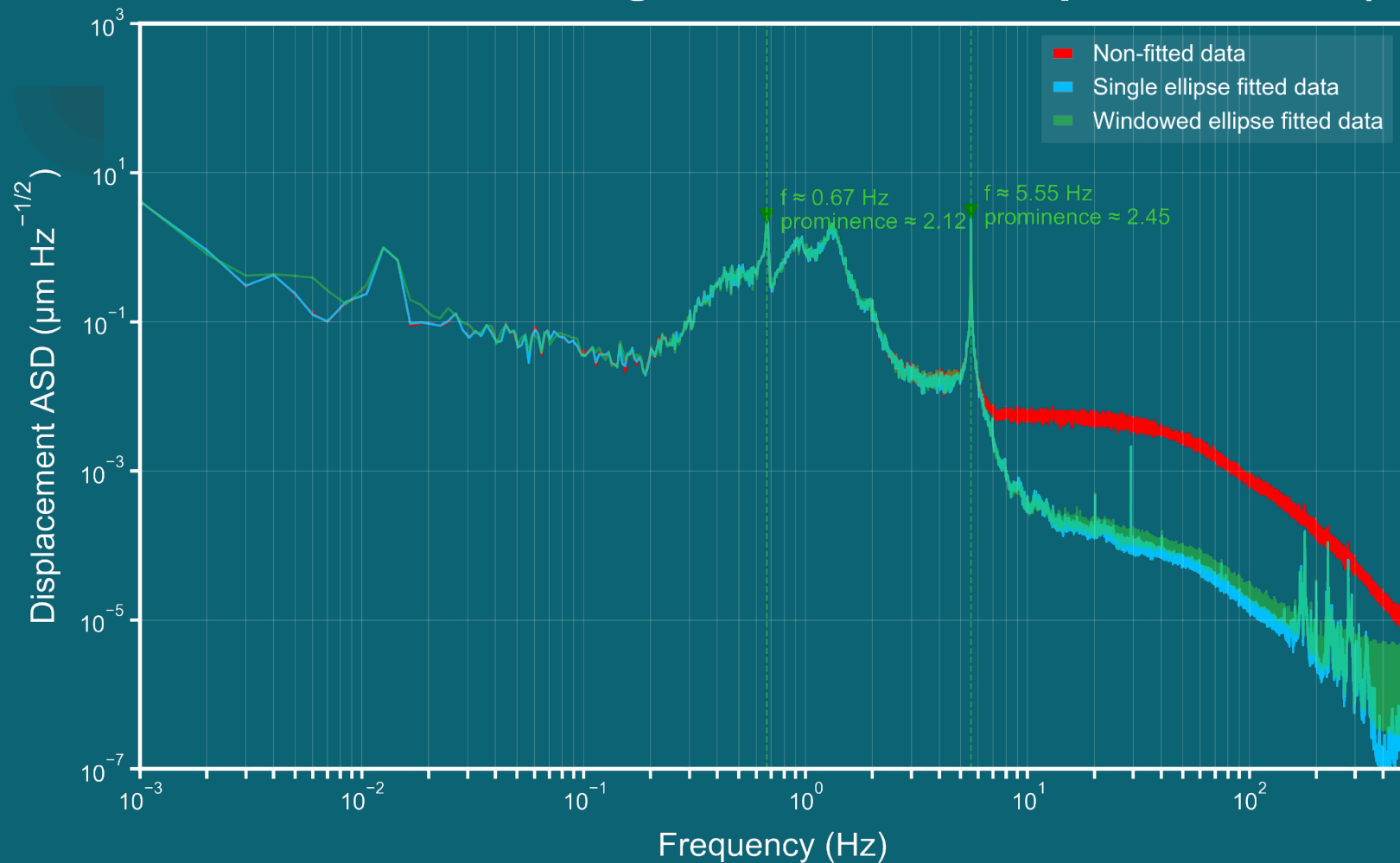
Smoothened ASD diagram for HoQI displacements (3x)



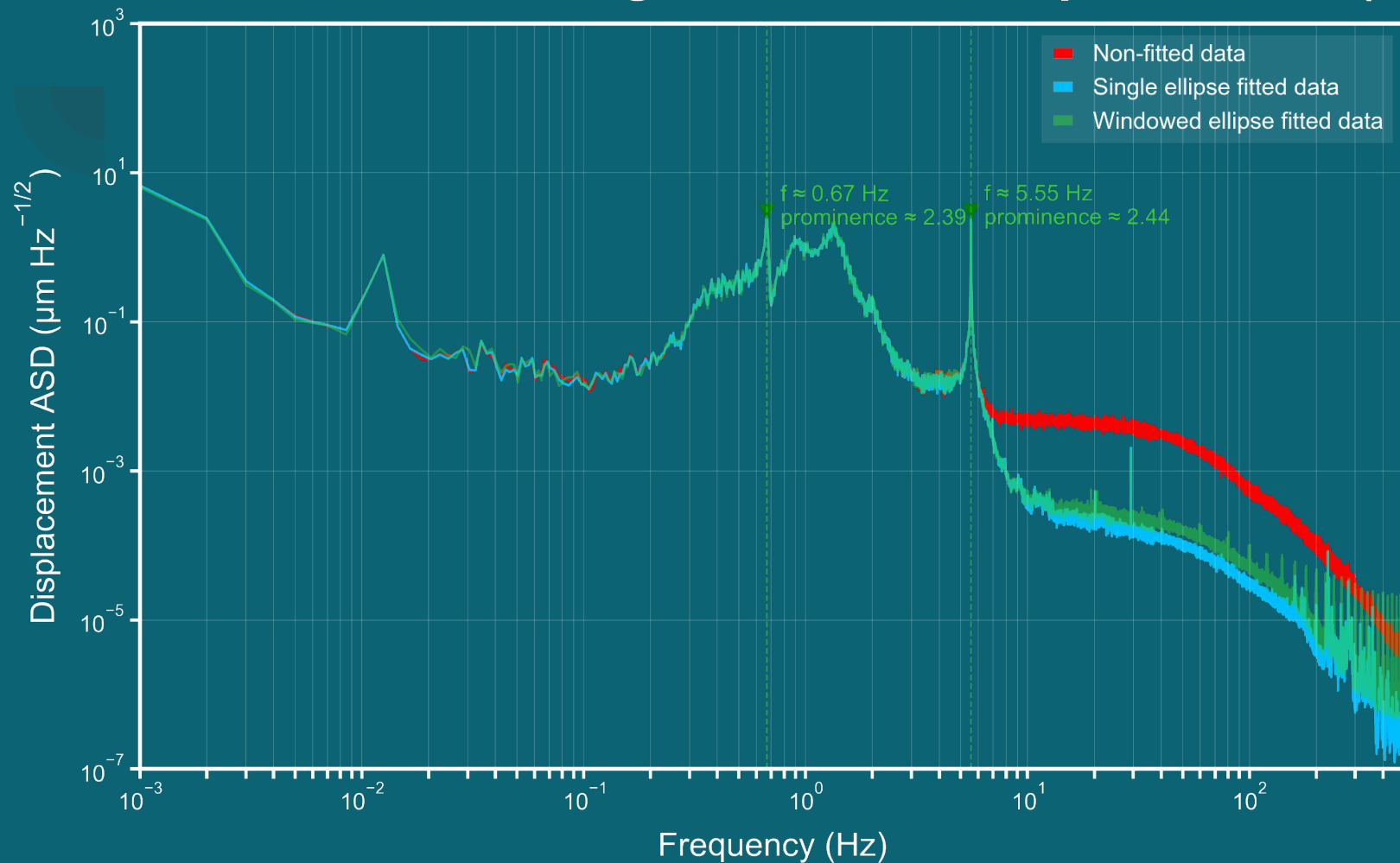
Smoothened ASD diagram for HoQI displacements (1z)



Smoothened ASD diagram for HoQI displacements (2z)



Smoothened ASD diagram for HoQI displacements (3z)





Real-time processing

Why?



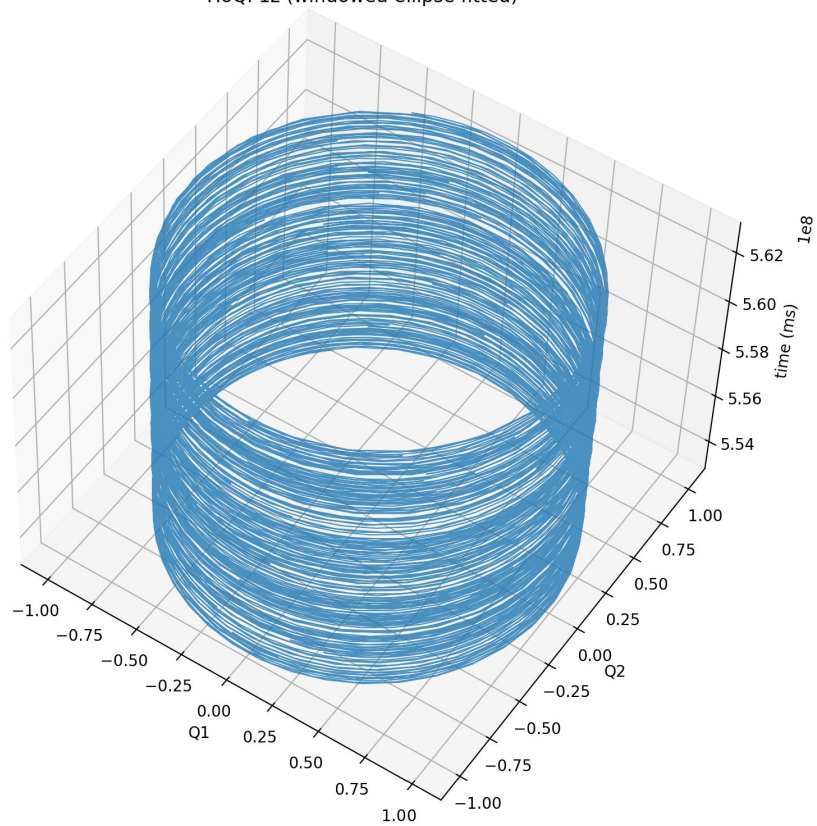
- Better results **during** measurements
- **Less** data needed
- Predicting **non-linear** effects → **correcting** real-time



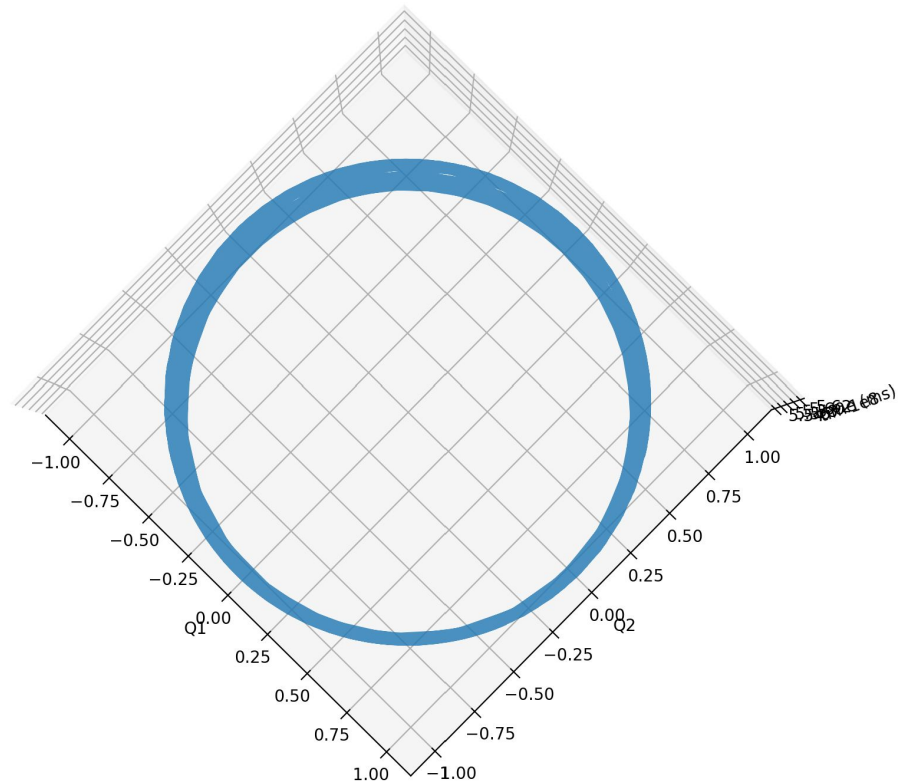
Prediction models

- Dependency orthogonal movement
- 3d plots
- fitting function or ML predictions

HoQI 1z (windowed ellipse fitted)



HoQI 1z (windowed ellipse fitted)

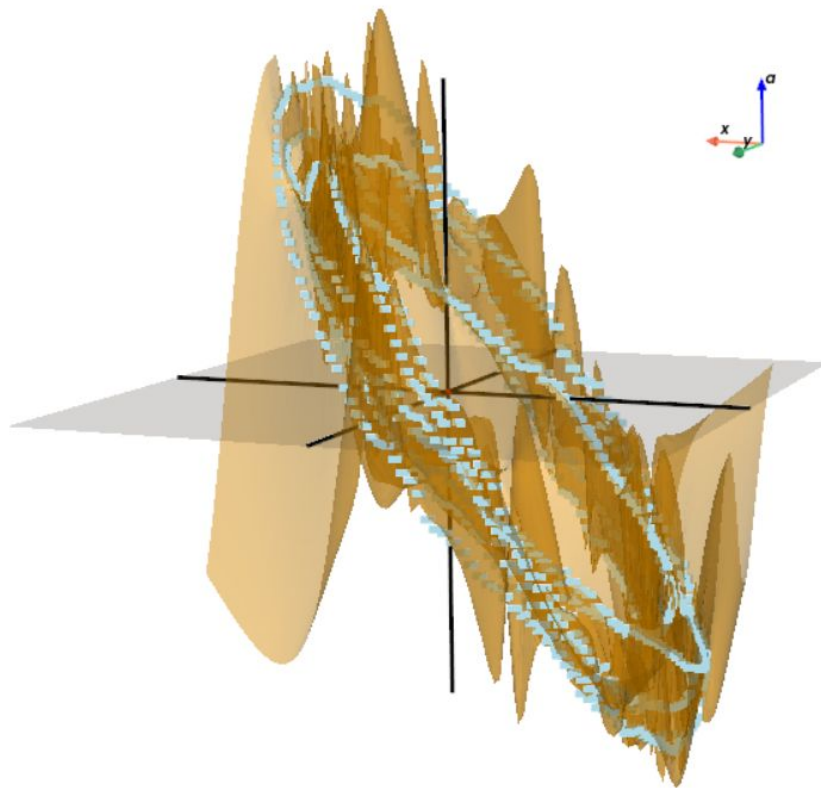




RBF

- 300 **chunks** (chunk size = 10,000)
- 9,901 **windows** per chunk → ellipse parameters
 - window size = 100
 - step size = 1
 - lag = 0
- Stored as **training data** in 'RBFInterpolator'
- **Local** RBF for first 9,901 points of the next chunk
 - neighbors = 100 (faster than global RBF)
 - smoothing = 0.1 (rough average → one-to-one)
- RBF used to **predict** the ellipse parameters
 - transformation with own, local RBF
 - (chunk 0 doesn't get transformed)

Ellipse parameter v/s position
in the orthogonal plane





kNN

Model parameters

- k
- Training size
- Memory
- Centre shift

kNN HoQI model results

RMSE lower is better

HoQI 1x

Active wins

k=100, uniform, p=2 | 11 blocks × 10,000

train: passive 80k · active 40k · gap 500

Active RMSE

0.00272

60.8% lower vs baseline

Base

0.00695 ± 0.00325

Passive

0.00302 ± 0.00063

Active

0.00272 ± 0.00104

56.6% lower vs baseline (passive)

Active vs passive: 9.8% lower than passive · better in 81.8% of blocks

Per parameter active change: x0 -9.9% · y0 -3.2% · a +30.6% · b +12.9% · θ +6.1%

HoQI 2x

Active wins

k=200, uniform, p=1 | 11 blocks × 10,000

train: passive 10k · active 20k · gap 500

Active RMSE

0.04924

3.4% higher vs baseline

Base

0.04761 ± 0.02725

Passive

0.05651 ± 0.02585

Active

0.04924 ± 0.02709

18.7% higher vs baseline (passive)

Active vs passive: 12.9% lower than passive · better in 81.8% of blocks

Per parameter active change: x0 +22.9% · y0 +21.0% · a +52.1% · b +10.6% · θ +12.1%

HoQI 3x

Active wins

k=500, distance, p=1 | 11 blocks × 10,000

train: passive 80k · active 80k · gap 500

Active RMSE

0.01426

5.8% higher vs baseline

Base

0.01348 ± 0.01303

Passive

0.01496 ± 0.01105

Active

0.01426 ± 0.01316

11.0% higher vs baseline (passive)

Active vs passive: 4.7% lower than passive · better in 72.7% of blocks

Per parameter active change: x0 +4.5% · y0 +4.0% · a +11.5% · b +7.6% · θ +4.5%

HoQI 1z

Passive wins

k=100, uniform, p=2 | 11 blocks × 10,000

train: passive 80k · active 80k · gap 500

Active RMSE

0.00352

65.8% lower vs baseline

Base

0.01029 ± 0.00443

Passive

0.00325 ± 0.00054

Active

0.00352 ± 0.00122

68.5% lower vs baseline (passive)

Active vs passive: 8.3% higher than passive · better in 54.5% of blocks

Per parameter active change: x0 +5.8% · y0 +14.4% · a -23.2% · b -21.5% · θ +3.6%

kNN HoQI model results

RMSE lower is better

HoQI 2z

Active wins

k=200, uniform, p=2 | 11 blocks × 10,000

train: passive 80k · active 40k · gap 500

Active RMSE

0.00903

31.7% lower vs baseline

Base

0.01323 ± 0.00378

Passive

0.00988 ± 0.00162

Active

0.00903 ± 0.00258

25.3% lower vs baseline (passive)

Active vs passive: 8.6% lower than passive · better in 63.6% of blocks

Per parameter active change: x0 +6.7% · y0 +7.9% · a +9.7% · b +4.6% · θ +9.7%

HoQI 3z

Active wins

k=200, uniform, p=2 | 11 blocks × 10,000

train: passive 20k · active 10k · gap 500

Active RMSE

0.02821

14.6% lower vs baseline

Base

0.03302 ± 0.00827

Passive

0.03346 ± 0.00483

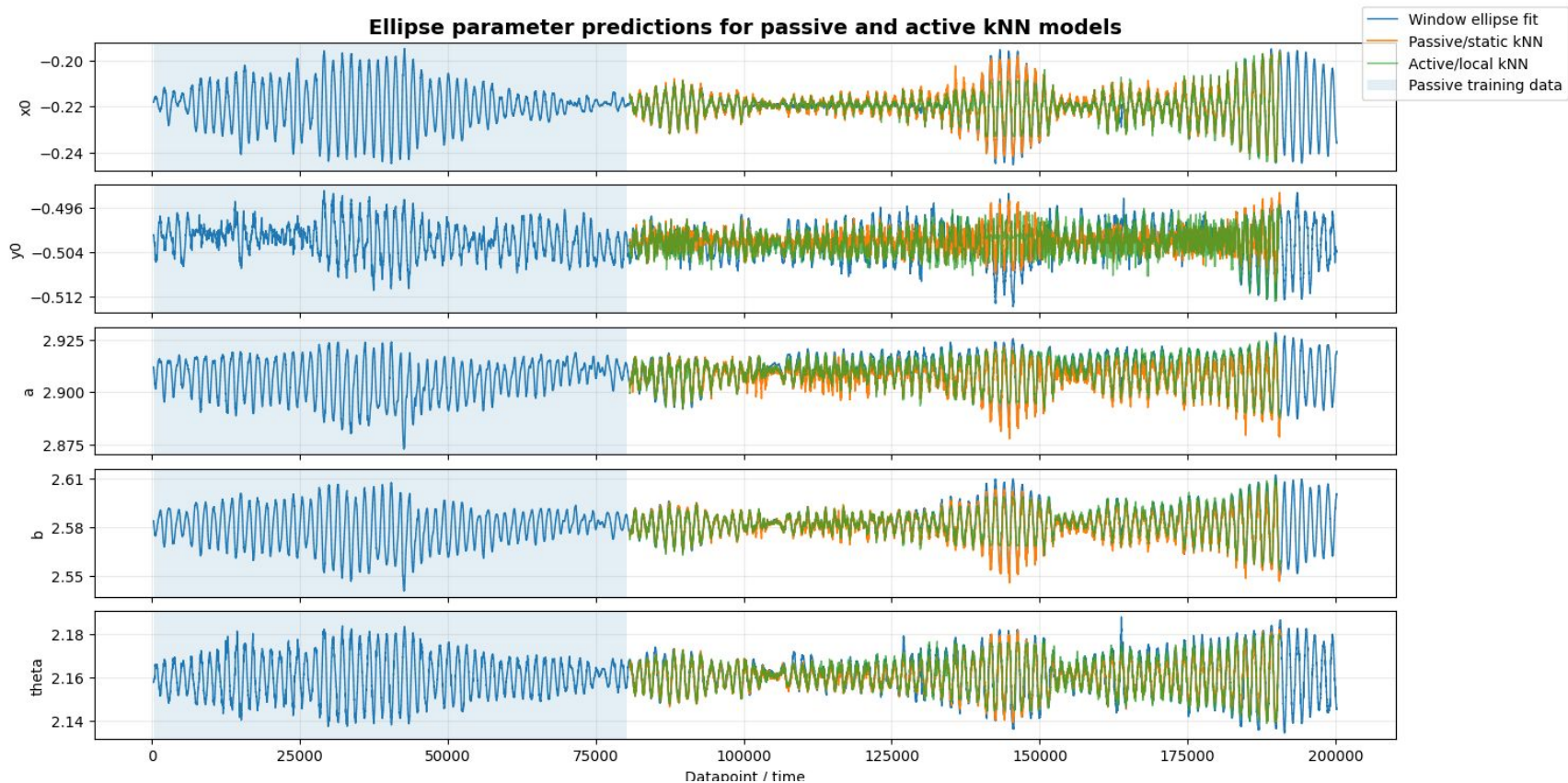
Active

0.02821 ± 0.00509

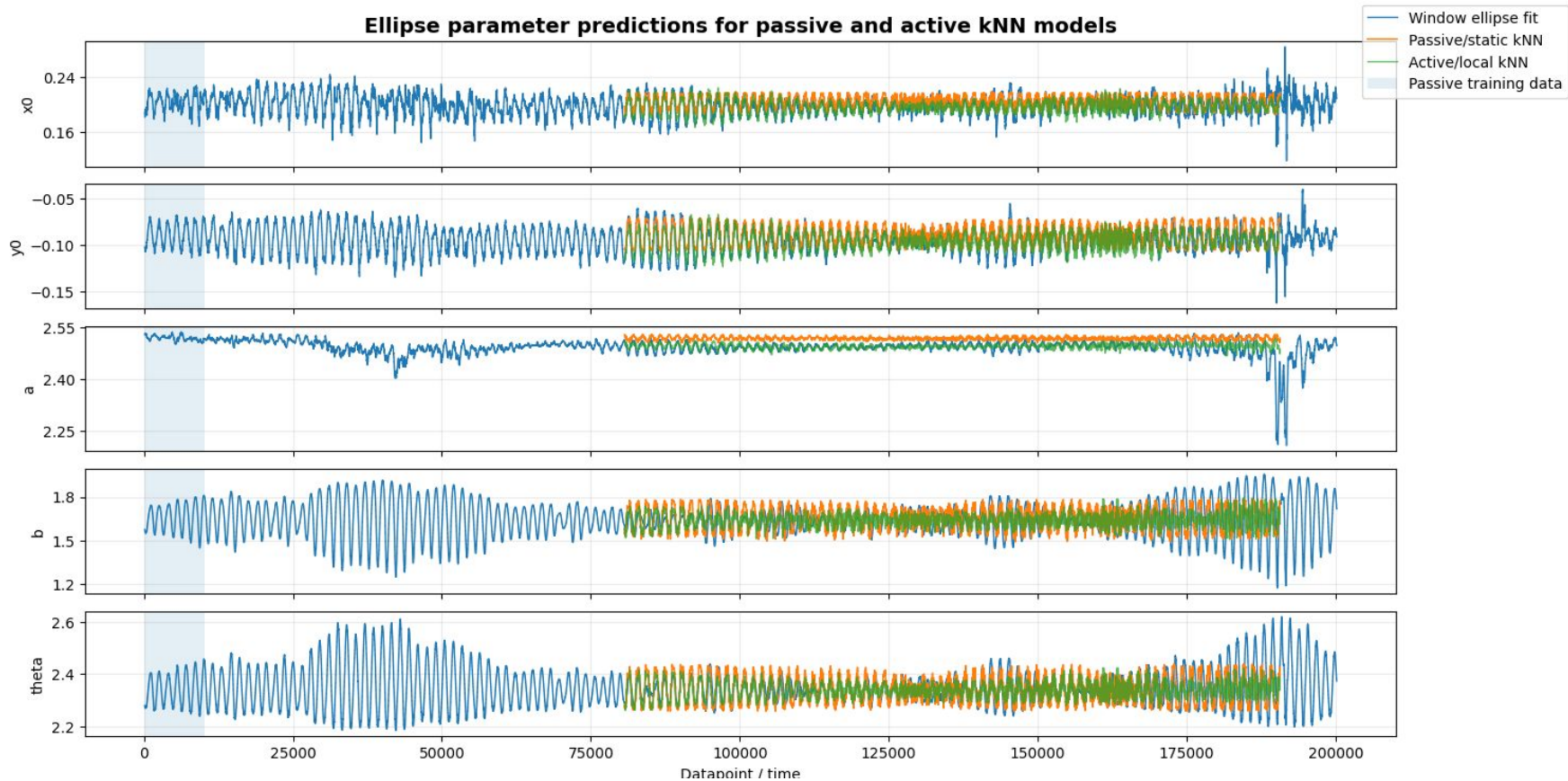
1.3% higher vs baseline (passive)

Active vs passive: 15.7% lower than passive · better in 81.8% of blocks

Per parameter active change: x0 +13.2% · y0 +20.0% · a +13.3% · b +10.4% · θ +15.6%

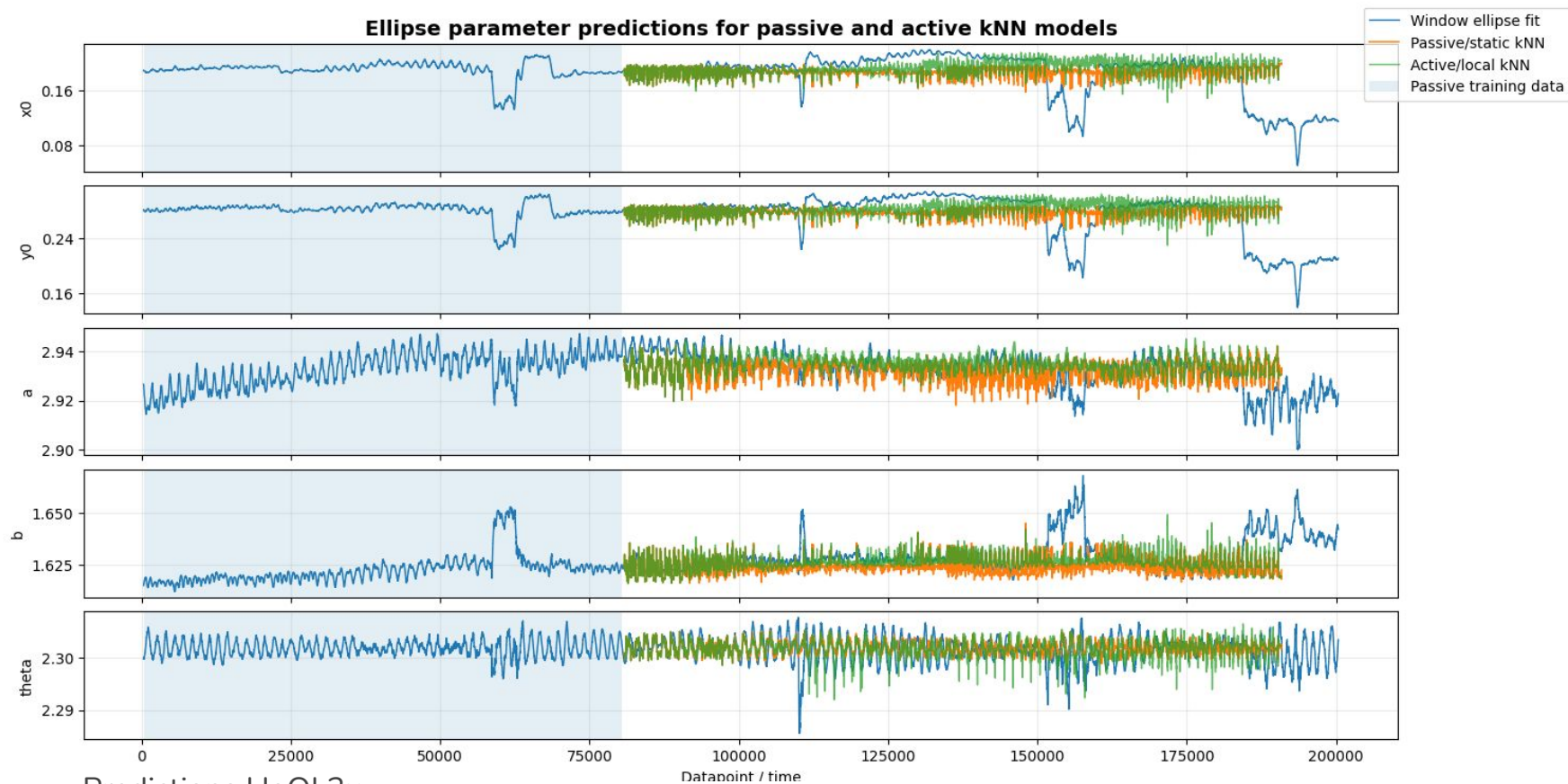


Predictions HoQI 1x

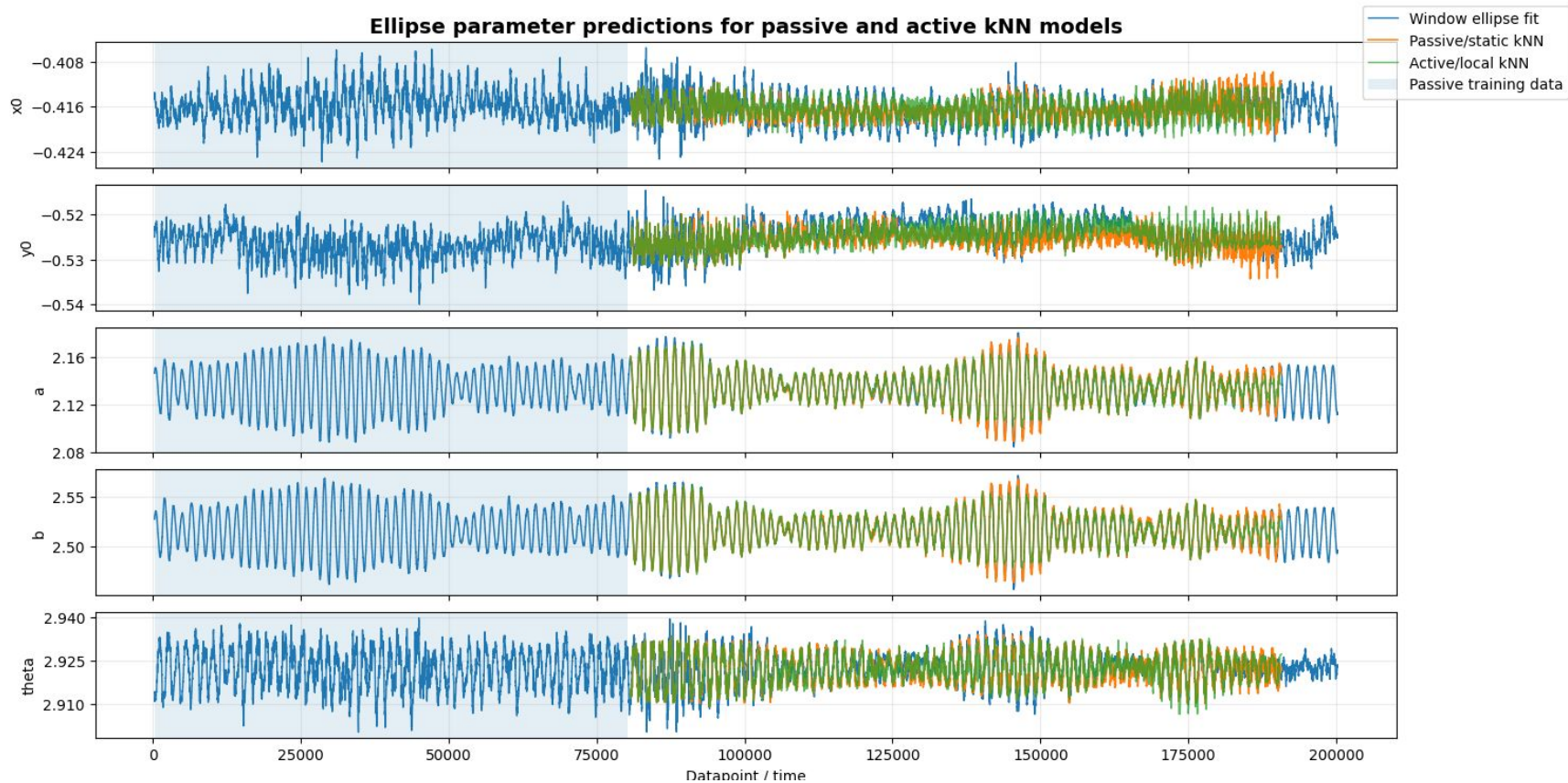


Predictions HoQI 2x

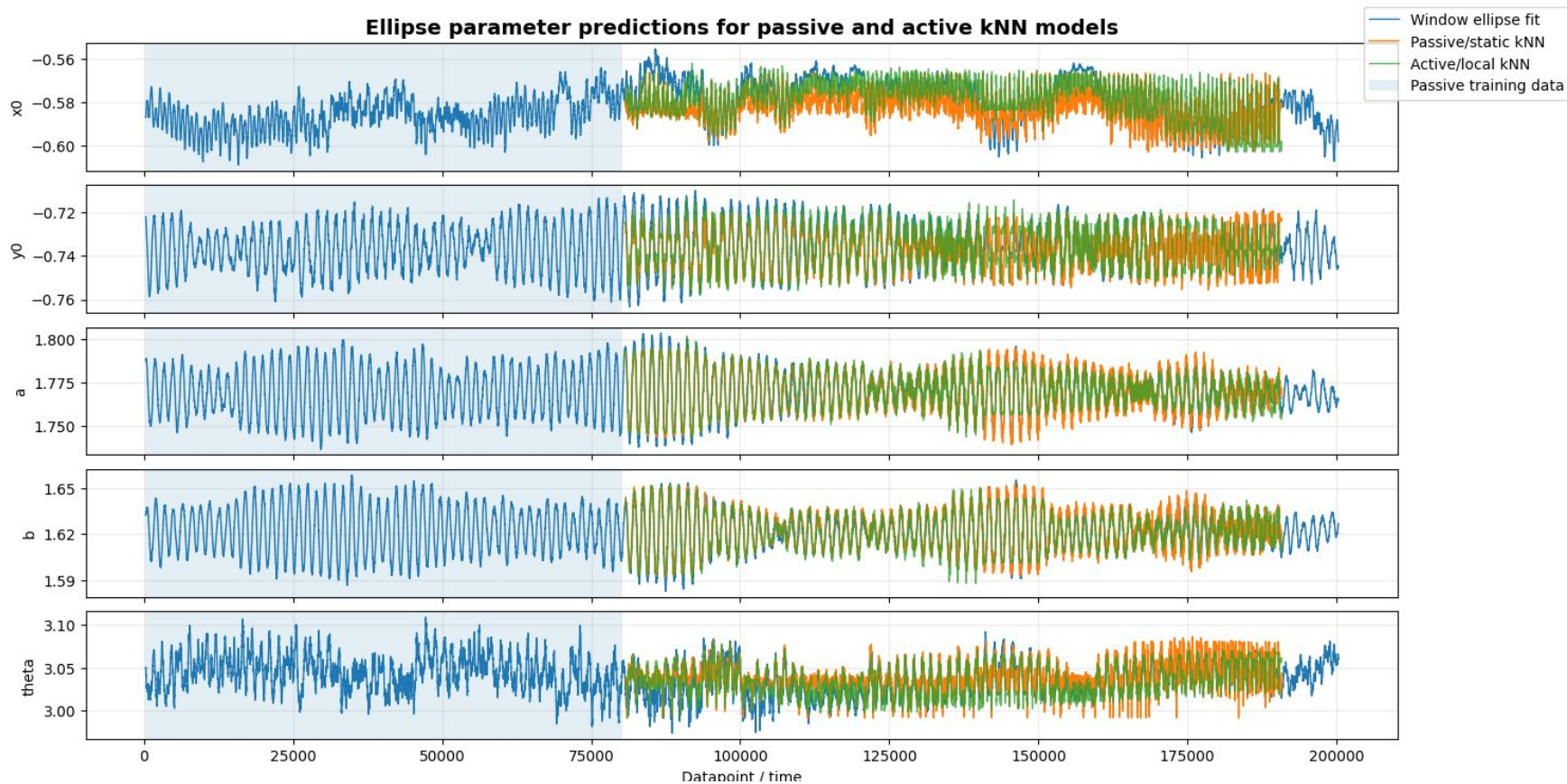
Ellipse parameter predictions for passive and active kNN models



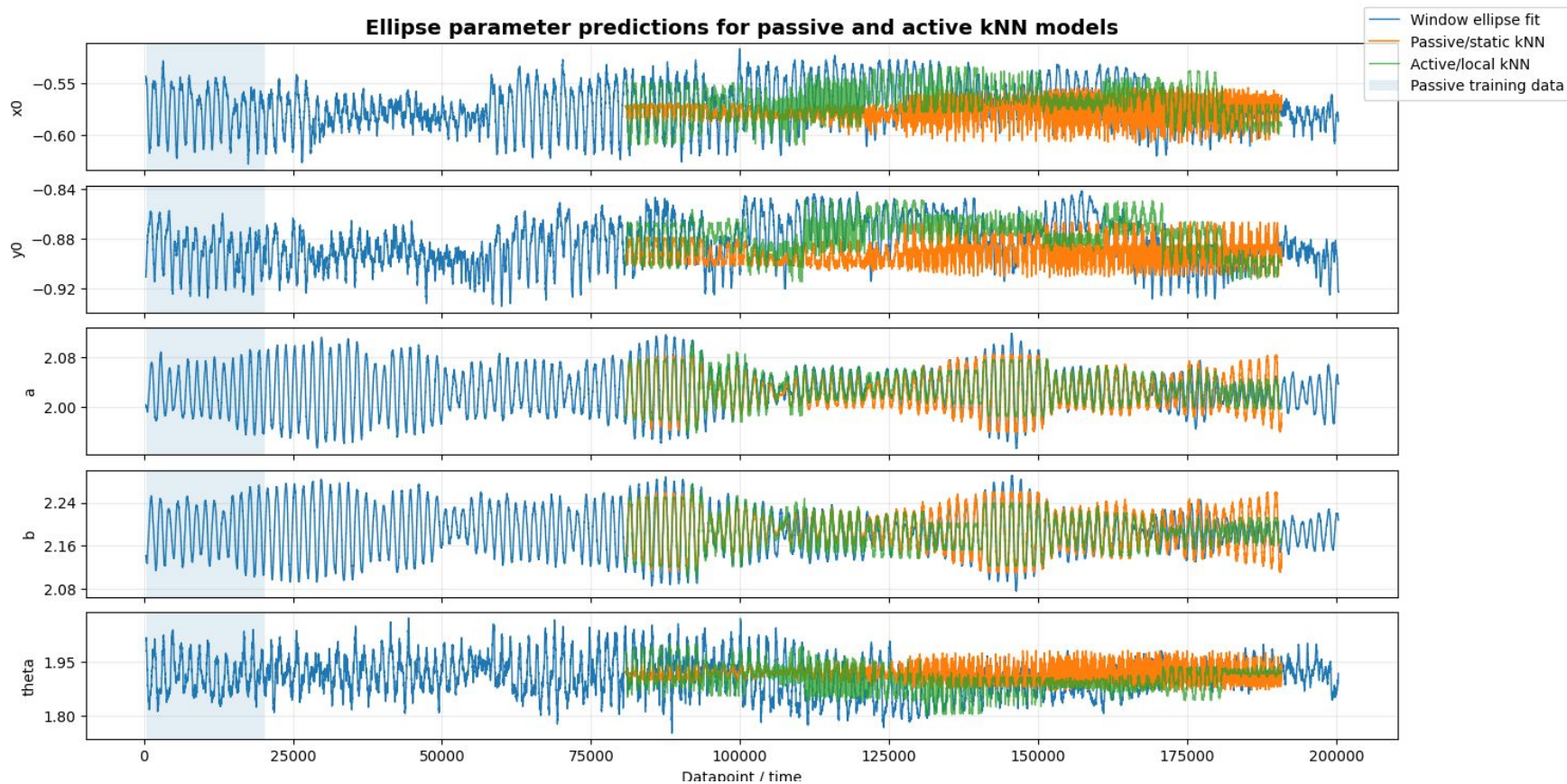
Predictions HoQI 3x



Predictions HoQI 1z



Predictions HoQI 1z



Predictions HoQI 1z



End Product Code





REDUCING NON-LINEARITIES IN HoQI BASED MEASUREMENTS



Introduction

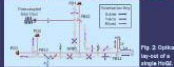
Gravitational waves are infinitesimal ripples in the fabric of spacetime. Predicted by Albert Einstein in 1916, their stretching and squeezing of spacetime as they pass by are detected using large laser interferometers, often characterized by their resonant L-shaped. Since they are transverse waves, gravitational waves stretch spacetime in one direction *perpendicular* to the propagation direction, while squeezing it in the other orthogonal direction. As a wave passes through the detector, the lengths of the detector's arms change slightly, causing a *phase shift* between two interfering laser beams. The resulting *interference pattern* implies the passing of a gravitational wave, whose propagation direction can be determined using *triangulation*.

Detecting gravitational waves requires measuring *subatomic fluctuations* in distance, demanding near-perfect mechanical isolation of the optical elements. To achieve this, external vibrations are constantly monitored by an active damping system of sensors and actuators, operating in tandem with passive isolators to suppress residual motion.

Method

A prototype of the aforementioned damping system is illustrated in Fig. 1 (1). The setup consists of three arms, all at an angle of roughly 134 degrees to each other. To each of the three sensors, two *homodyne* (homodyne interferometers) are attached. These HoQIs serve as the high-precision sensors responsible for measuring the system's resonances. They use *laser interferometry* to measure the position of a free-floating test mass in the plane perpendicular to the associated arm. An optical beat signal of a single HoQI can be seen in Fig. 2 (2). The three photodetectors (PD1, PD2 and PD3) measure the intensity of the interfering light, converting it to a voltage used to calculate the position of the test mass.

Combining the individual measurements of all six HoQIs, we can use the *6x6 transformation matrix* as given in Eq. (1), to determine the *displacements* of the detector's center of mass (CM) in all six degrees of freedom (X, Y, Z, α , β , γ).



$$\begin{pmatrix} -\frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & \frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Time series of 6x6 displacements (unwindowed ellipse fitted).

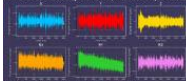


Fig. 3a Time series of the CM position in all six degrees of freedom (unwindowed ellipse fitted).

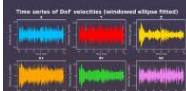


Fig. 3b Time series of the CM velocity in all six degrees of freedom (unwindowed ellipse fitted).

Time Series

Using this transformation matrix, we are able to make a so-called *time series* of both the displacement (see Fig. 3a) and the velocity (see Fig. 3b) of the center of mass in all six degrees of freedom. The data used to create these plots come from a 3666-second measurement, with a fixed sample rate of exactly 1000 Hz.

These time series can be used to gain insight into the movements of the detector. As will be shown later, this is crucial for the efficient removal of non-linearities from the dataset. *Non-linearities* are systematic distortions where the output signal is no longer directly proportional to the physical input. They are usually caused by either high-amplitude or high-velocity movements, resulting in the situation of *instrumental noise*.

Improving Linearity

There are multiple ways in which non-linearities could get injected into the measurements. For instance, if the beam splitters in Fig. 2 do not split the laser beams in a perfect 50/50 ratio, the photodetectors measure differing intensities, leading to erroneous results. Furthermore, the quarter-wave plate might not delay the oscillation in one direction with exactly π degrees, resulting in a non-perfect shift of π . A final cause for non-linearities is the potential *time dependence* of the fringe visibility. *Fringe visibility* is a quantitative measure of the contrast of interference fringes in any wave-based system. Possible reasons of a time-dependent fringe visibility include imperfect selection mirrors, a non-perfect 50/50 ratio of the beam splitters, a mechanical flow in the half-wave plate, and movements in a HoQI's transverse plane.

This research applied *ellipse fitting* to reduce the non-linearities in HoQI-based measurements, with a specific focus on the *high-frequency domain*. We have done this using various types of data analysis, including ellipse fitting, ASD analysis, and training a machine learning model with the usage of a KNN algorithm.

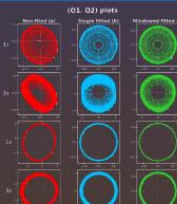


Fig. 4 Q1-Q2 plots for four HoQIs (3a, 3b, 3c and 3d) comparing (a) the unwindowed, (b) the single ellipse fitted, and (c) the windowed ellipse fitted data.

Ellipse Fitting

The first analytical method we used for reducing the injected non-linearities, is called *ellipse fitting*. As can be seen in the *animation* [scan the QR code to the top left corner], plotting the Q1 fit of a certain HoQI against the Q2 fit of that same HoQI, should - in theory - result in a perfect circle. Though, existing non-linearities in the measurements cause the data points to fall off this circle, leading to a Lissajous-like figure. In practice, however, the (Q1, Q2) points make take on the appearance of an ellipse (see Fig. 4a).

The aforementioned ellipse fitting is a process where we fit one or more ellipses through the plotted data points, after which we use the determined ellipse parameters (a, b, α , β and γ) to map the data points back onto a single circle - the *unit circle*. In our case, An image of the results corresponding to two different ways of ellipse fitting (called *single ellipse fitting* and *windowed ellipse fitting*) can be seen in Fig. 4b and Fig. 4c respectively.

During the analysis, we also constructed a *time series* of all the ellipse parameters. This revealed that the values of these parameters *fluctuate* over time. Fitting a single ellipse through the (Q1, Q2) data points means every point is *transformed* with the *exact same parameters*, leading to slightly incorrect transformation. Windowed ellipse fitting tries making up for this by fitting an ellipse through the data points for numerous 'windows', using an adjustable window size, step size, and lag. Consequently, each point is transformed using locally optimized parameters, yielding more accurate results.

ASD Plotting

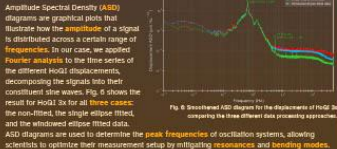


Fig. 5 Broadband ASD diagram for the displacements of HoQI 3a, converting the time series of the different data processing approaches.

As shown in the (Q1, Q2) plots in Fig. 4, applying ellipse fitting methods yields significant improvements in linearity compared to the non-fitted data. While the quality increase relative to single ellipse fitting is sometimes *minimal*, windowed ellipse fitting offers the best overall results.

KNN Algorithm

One of our hypotheses was that the ellipse parameters depend on the position of the test mass in the transverse plane of the relevant HoQI. To test this hypothesis, a KNN algorithm was used for training a *machine learning model* to check whether it could accurately predict future parameters (see Fig. 7). We compared a static KNN model, whose training data consists of the first 88,000 data points, to an active KNN model, whose training data consists of constantly updating data points.

Both KNN models mostly achieved RMSE values below the baseline (see Tab. 1), indicating that the ellipse parameters can indeed be predicted from the position in the transverse plane. The active model performed slightly better than the static model for most of the data segments we tested them on. This suggests that the underlying patterns remain relatively stable over time, although temporal non-linearities may still be present.

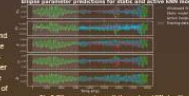


Fig. 7 Ellipse parameter predictions using a KNN algorithm, trained for both an active and a static model.

	RMSE (a.u.)	Gate (%)
Baseline	6.702 ± 0.725	-
Static	2.015 ± 0.122	20.7
Active	2.001 ± 0.118	20.9
Tab. 1 RMSE results. Gate relates to baseline.		

The ASD diagram, as displayed in Fig. 5, further reinforces this statement. Especially in the high-frequency domain, a substantial reduction in noise levels has been achieved. This has led to more clearly visible peak frequencies, thereby enabling the precise detection and subsequent removal of, for example, specific bending modes. Finally, the KNN results, as presented in Fig. 7 and Tab. 1, confirm our hypothesis regarding the dependence of the ellipse parameters on the test mass' position in the transverse plane. This dependence is subject to minor temporal fluctuations, driven by underlying, independently varying non-linearities.

References

- [1] A. L. Michel, 'HoQI Low-noise Interferometric Sensors for the Future of Gravitational Wave Detection', PhD thesis, Department of Physics and Astronomy, Vrije Universiteit Amsterdam (2024).
- [2] S. A. Cooper et al., 'A compact, large-range interferometer for precision measurement and length sensing', in *Classical and Quantum Gravity* 20 (2003), no. 10, 010001-1-010001-6.

ANIMATION TIME

